

Aisha Adam Opaluwa

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AI Engineer & Researcher

I build AI systems that can be evaluated, explained, and trusted. My work spans agentic LLM systems with formal state control, RAG pipelines with rigorous failure analysis, LLM evaluation benchmarking, mechanistic interpretability, and privacy-preserving ML. I care equally about building things that work and measuring honestly when they do not.

EDUCATION

Carnegie Mellon University

MSc, Engineering & Artificial Intelligence

Aug 2024 – Dec 2025

- Teaching Assistant: Introduction to Deep Learning (Spring 2025)
- Teaching Assistant: Research Methods in Engineering (Fall 2025)

Federal University of Technology, Minna

B.Eng, Computer Engineering

Sep 2015 – Sep 2021

PROJECTS

Quilltale: Agentic LLM System with Formal World State | github | live

2026

- Separated story generation (LLM) from world state enforcement (formal Python dataclass model) — the LLM cannot override tracked facts about locations, items, or NPC state
- Implemented validated state transitions: invalid actions caught, logged, and reflected in narration — a concrete testbed for LLM controllability over unbounded sessions
- Built per-NPC episodic memory with significance scoring; past interactions retrieved and injected into context for consistent agent behaviour regardless of session length
- Developed LLM-as-judge evaluation framework (20-turn fixed scenario); factual consistency 85%, memory utilisation 91.7%, invalid transition rate ~12%; failure analysis identified three distinct failure modes including plausible invention and narration-state lag

Scientific Paper QA: RAG Pipeline with Systematic Failure Analysis | github | live

2026

- End-to-end RAG pipeline (Llama 3.2, LlamaIndex, ChromaDB); 156 chunks, semantic embedding via BAAI/bge-small-en-v1.5; fully local via Ollama with no API cost
- Designed 34-question evaluation dataset with expert-verified answers; hybrid keyword-number-acronym scorer; top-k from 3→5 improved accuracy from 44.1% to 50%
- Failure analysis explicitly distinguishes retrieval misses from model hallucination — two failure modes with different remedies; documented residual failures including cross-paper ambiguity

SciReason-Bench: LLM Scientific Reasoning Benchmark | github | live

2026

- 100-question benchmark evaluating 4 LLMs (Gemini, Phi-3, Mistral, Llama) across 5 categories: factual recall, conceptual explanation, numerical reasoning, cross-domain synthesis, and calibration/uncertainty
- Key finding: model size does not predict performance — Phi-3 (3.8B) outperforms Mistral (7B); numerical reasoning is the sharpest open-source/commercial gap (36–45% vs 95%)
- Documented self-preference bias when the evaluated model family serves as judge — a methodological limitation with direct implications for reliable agentic evaluation pipelines

arXiv Abstract Classifier: Fine-tuning & Mechanistic Interpretability | github | live

2026

- Fine-tuned DistilBERT on 28k arXiv abstracts across 11 fields; trained on PSC V100 via SLURM; 88% test accuracy, 0.877 macro F1; tracked with WandB; model published to HuggingFace Hub
- Captum integrated gradients for token-level explainability; identified semantically meaningful confusion (cs.AI ↔ cs.NE); OOD analysis on materials science abstracts with detection proposed as mitigation

Quantum Materials Research | Manuscript in Preparation

Summer 2025 – Present

- DFT simulations on wolframite oxides using Quantum ESPRESSO on HPC clusters (PSC Bridges-2); investigating band gap narrowing and conductivity trends

Monkeypox Detection with Explainability | view

Spring 2025

- Extended published VGG16 model with ConvNeXt transfer learning; multi-fold evaluation to 88.9% accuracy on best fold; Grad-CAM for clinical interpretability

Multimodal Emotion Recognition for ASD Support | view

Fall 2024

- Fused CNN facial expression analysis with LSTM speech emotion recognition (MFCC); empirically validated audio-visual fusion outperforms unimodal baselines

Chatbot Access Advantage Study (QCRI)

Fall 2025

- Human-AI interaction study with Qatar Computing Research Institute; analysed student-AI dialogues to measure how modality (text/audio), mobile access, and translation affect engagement and dialogue complexity

AI Governance: GDPR vs African Union Data Policy Framework

Fall 2025

- Comparative policy analysis on enforcement mechanisms and AI deployment implications across African jurisdictions; proposed governance recommendations for strengthening AI-relevant data frameworks

Speech & Language Processing (Course Projects)

Spring 2025

- ASR via BiLSTM seq2seq with CTC loss; Transformer encoder-decoder with beam search; evaluation using Levenshtein distance and CER

Computer Vision & Signal Processing (Course Projects)

Spring 2025

- CNN face verification for identity matching; phoneme recognition with MFCC extraction and MLP classifiers

PROFESSIONAL EXPERIENCE

Research Assistant

Carnegie Mellon University

Jan 2026 – Present

Software Engineer

Recherché Inc

Mar 2021 – Dec 2025

New York, USA (remote)

- **Project Guardian (MAS)** — led frontend engineering and contributed to smart contract integration and backend for the Monetary Authority of Singapore's institutional DeFi pilot alongside JP Morgan, DBS, and SBI; tokenized real-world asset lending platform (MAS announcement)
- **Institutional DiD** — built decentralised identity components (frontend, mobile apps, SDKs, smart contracts) for a major investment bank as part of a project that resulted in two granted patents on distributed ledger identity management
- **Monadic DNA** (monadicdna.com) — led frontend, iOS and Android app development; built the Rust library service enabling cross-platform cryptographic computation (compiled for both iOS and Android); contributed to backend and Nillion MPC integration; built SnipperBot end-to-end: parses 23andMe SNP data, stores secrets on Nillion's MPC network, runs blind trait analysis without raw data ever being decrypted, issues on-chain VerifiedTrait attestations via Sign Protocol; deployed with 30+ real participants using live genomic data
- Full-stack across TezFin DeFi lending protocol, TEZEX exchange, and Galleon wallet on Tezos mainnet; smart contract integration and blockchain operations across all three
- **Persimmon** (App Store) — led end-to-end development of a live iOS app for privacy-focused crypto wallet monitoring; daily balance alerts for self-hosted wallets; shipped to production on the Apple App Store

Data Analyst

Innovate for Africa

Jan 2021 – Feb 2021

Boston, United States

- Predictive modelling (Linear Regression, Random Forests) for urban housing analytics

Frontend Developer

TIIDelab

Jul 2020 – Dec 2020

- Built a React timesheet application with responsive design; led volunteer tech initiatives supporting internally displaced persons

SKILLS

Languages: Python, Rust, C++, TypeScript, JavaScript

AI & ML: PyTorch, HuggingFace Transformers, LlamaIndex, ChromaDB, Captum, WandB, Ollama, scikit-learn, Pandas, NumPy, Matplotlib, Gemini API

Infrastructure: HPC/SLURM (PSC Bridges-2, TRACE), Quantum ESPRESSO, Docker, Git, Streamlit, Gradio, Flask

Web & Frameworks: React, Redux, GraphQL, SwiftUI, Web3.js, Ethers

Research Areas: Agentic AI Systems, LLM Evaluation & Benchmarking, RAG Systems, Mechanistic Interpretability, Human-AI Interaction, AI Governance & Policy, Scientific ML, Privacy-Preserving ML (FHE/MPC)

LEADERSHIP & VOLUNTEERING

Community Tech Leadership: Coordinated Google Developers Group (GDG) and Women Techmakers (WTM) programs engaging 200+ community members

Innovate Research Academy: Led sessions on research writing and technical communication

IDP Camp Project Manager: Surpassed fundraising goals by 35% in 48 hours, supporting renovations for 200+ displaced persons